



# **DOMESTIC REFRIGERATION AND AIR-CONDITIONING SERVICING**

**TECHNICAL-PROFESSIONAL**

## **Grade 11**

**Three-Term Budget Of Work (BOW)  
for Learning Competencies**

**TECHNICAL-PROFESSIONAL  
INDUSTRIAL TECHNOLOGIES  
Grade 11**

**Three-Term Budget Of Work  
For Learning Competencies**

*Please note that each Tech-Pro elective is completed in three terms if delivered at Grade 11.*

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| <b>Elective</b>     | <b>Domestic Refrigeration and Air-Conditioning Servicing</b> |
| <b>Prerequisite</b> | <b>None</b>                                                  |

| <b>1<sup>st</sup> TERM</b>  |                                                                                                                     |
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| <b>Performance Standard</b> | The learner performs the installation of domestic refrigeration and air-conditioning units with safety precautions. |
| <b>Content Standard</b>     | Understands the concepts and principles of installing domestic refrigeration and air-conditioning units.            |

| <b>Week</b> | <b>Learning Competency</b>                                                                                          | <b>Suggested Activities</b>                                                                                                                                                                                                                                                         |
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| <b>1</b>    | <ul style="list-style-type: none"> <li>Discuss overview of Refrigeration and Air Conditioning servicing.</li> </ul> | <ul style="list-style-type: none"> <li><b>Service Cycle Exploration</b> <ul style="list-style-type: none"> <li>Learners study the basic steps and sequence of refrigeration and air conditioning servicing, including maintenance and troubleshooting tasks.</li> </ul> </li> </ul> |

| Week | Learning Competency                                                                                                     | Suggested Activities                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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|      |                                                                                                                         | <ul style="list-style-type: none"> <li>○ Observe a demonstration or video of the service cycle, then identify and list the key steps involved.</li> <li>● <b>Servicing Tools Identification</b> <ul style="list-style-type: none"> <li>○ Learners identify common tools and equipment used in refrigeration and air conditioning servicing.</li> <li>○ Examine each tool, describe its purpose, and demonstrate correct handling.</li> </ul> </li> </ul>                                                                                                                                                                                                                                        |
| 2    | <ul style="list-style-type: none"> <li>● Discuss refrigeration and air conditioning concepts and principles.</li> </ul> | <ul style="list-style-type: none"> <li>● <b>Principle Mapping</b> <ul style="list-style-type: none"> <li>○ Learners explore key concepts like the refrigeration cycle, heat transfer, and thermodynamic principles used in air conditioning systems.</li> <li>○ Create a concept map showing components, flow of refrigerant, and energy transfer in a refrigeration system.</li> </ul> </li> <li>● <b>Case Study Analysis</b> <ul style="list-style-type: none"> <li>○ Learners analyze real-world air conditioning problems to understand practical applications of refrigeration principles.</li> <li>○ Review a case study of a malfunctioning AC system and explain</li> </ul> </li> </ul> |

| Week | Learning Competency                                                                                                     | Suggested Activities                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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|      |                                                                                                                         | which principles are involved in diagnosing the problem.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 3    | <ul style="list-style-type: none"> <li>Perform pre-installation procedure for an air-conditioning unit.</li> </ul>      | <ul style="list-style-type: none"> <li><b>Site Inspection Drill</b> <ul style="list-style-type: none"> <li>Learners assess the installation site for window or split-type AC units, checking measurements, power supply, and support structures.</li> <li>Inspect the designated site, measure openings, and verify electrical and structural requirements before installation.</li> </ul> </li> <li><b>Component Preparation</b> <ul style="list-style-type: none"> <li>Learners prepare AC unit components (filters, brackets, screws, electrical wires) for installation.</li> <li>Collect, organize, and inspect all AC components to ensure readiness for installation.</li> </ul> </li> </ul> |
| 4    | <ul style="list-style-type: none"> <li>Perform procedures in installing a window-type air-conditioning unit.</li> </ul> | <ul style="list-style-type: none"> <li><b>Mounting Practice</b> <ul style="list-style-type: none"> <li>Learners practice mounting a window-type AC unit securely in a frame or window opening.</li> <li>Position the AC unit in the window opening, level it, and secure it using brackets and screws.</li> </ul> </li> <li><b>Electrical and Functional Check</b></li> </ul>                                                                                                                                                                                                                                                                                                                       |

| Week                               | Learning Competency                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Suggested Activities                                                                                                                                                                                                                                                         |
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|                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <ul style="list-style-type: none"> <li>○ Learners connect the AC unit to a power supply and verify correct operation, cooling, and safety features.</li> <li>○ Plug in the unit, turn it on, and check for proper cooling, airflow, and safe electrical function.</li> </ul> |
| <b>Suggested Performance Tasks</b> | <ul style="list-style-type: none"> <li>● <b>Individual Performance Task - Window-Type AC Installation</b> <ul style="list-style-type: none"> <li>○ Each learner independently installs a window-type air-conditioning unit, ensuring correct placement, secure mounting, proper electrical connection, and adherence to safety procedures.</li> <li>○ Measure and prepare the installation site, mount the unit securely, connect the power supply safely, and test for proper operation.</li> </ul> </li> <li>● <b>Group Performance Task - Domestic Refrigerator Setup Project</b> <ul style="list-style-type: none"> <li>○ In groups, learners collaboratively install a domestic refrigeration unit, dividing tasks such as positioning, leveling, electrical connection, and safety checks, ensuring proper installation and operation.</li> <li>○ Work as a team to position the refrigerator, connect power safely, verify leveling and clearance, and test all functions according to safety standards.</li> </ul> </li> </ul> |                                                                                                                                                                                                                                                                              |

**2<sup>ND</sup> TERM**

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| <b>Performance Standard</b> | The learner performs the maintenance of domestic refrigeration and window-type air-conditioning units with safety precautions. |
| <b>Content Standard</b>     | Understands the concepts and principles of maintaining domestic refrigeration and window-type air-conditioning units.          |

| <b>Week</b>  | <b>Learning Competency</b>                                                                                   | <b>Suggested Activities</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| <b>1 - 5</b> | <ul style="list-style-type: none"> <li>Perform maintenance of window-type air-conditioning units.</li> </ul> | <ul style="list-style-type: none"> <li><b>Air Filter and Coil Cleaning</b> <ul style="list-style-type: none"> <li>Learners clean and inspect the air filters and evaporator/condenser coils to ensure proper airflow, efficiency, and unit longevity.</li> <li>Remove the filters and coils, clean them using water, brushes, or compressed air, inspect for damage, and reinstall correctly.</li> </ul> </li> <li><b>Electrical and Drainage System Check</b> <ul style="list-style-type: none"> <li>Learners inspect the unit's electrical connections, controls, and drainage system to ensure safe operation and proper water flow.</li> <li>Check wiring and terminals, test fan and thermostat functions, and clear the drain pan and pipe of debris.</li> </ul> </li> </ul> |

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| <b>Suggested Performance Tasks</b> | <ul style="list-style-type: none"> <li>• <b>Individual Performance Task - Routine AC and Refrigerator Maintenance</b> <ul style="list-style-type: none"> <li>○ Each learner independently performs routine maintenance on a domestic refrigerator or window-type AC unit, including cleaning filters, coils, and drainage, and checking electrical connections safely.</li> <li>○ Follow a step-by-step maintenance procedure, clean and inspect all components, and ensure proper operation while observing safety precautions.</li> </ul> </li> <li>• <b>Group Performance Task - Comprehensive Maintenance Workshop</b> <ul style="list-style-type: none"> <li>○ In groups, learners collaboratively perform full maintenance on multiple refrigeration and AC units, dividing tasks such as coil cleaning, filter replacement, electrical inspection, and drainage verification, while following safety protocols.</li> <li>○ Work as a team to perform thorough maintenance on assigned units, checking all mechanical and electrical systems, and confirm proper operation safely.</li> </ul> </li> </ul> |
| <b>Content Standard</b>            | Understands the concepts and principles of troubleshooting domestic refrigeration and window-type air-conditioning units.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Performance Standard</b>        | The learner performs troubleshooting of domestic refrigeration and window-type air-conditioning units with safety precautions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

| <b>Week</b>  | <b>Learning Competency</b>                                                                                                                        | <b>Suggested Activities</b>                                                                                                                                                                                                                                                                                                                                                                                                         |
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| <b>6 - 7</b> | <ul style="list-style-type: none"> <li>• Discuss pre-troubleshooting of domestic refrigeration and window-type air conditioning units.</li> </ul> | <ul style="list-style-type: none"> <li>• <b>Common Fault Identification</b> <ul style="list-style-type: none"> <li>○ Learners study typical problems in domestic refrigerators and window-type AC units, such as poor cooling, strange noises, leaks, and electrical issues.</li> <li>○ Examine demonstration units or case examples and list common faults, their possible causes, and observable symptoms.</li> </ul> </li> </ul> |

| Week   | Learning Competency                                                                                                                           | Suggested Activities                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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|        |                                                                                                                                               | <ul style="list-style-type: none"> <li>● <b>Troubleshooting Checklist Creation</b> <ul style="list-style-type: none"> <li>○ Learners develop a pre-troubleshooting checklist to guide systematic inspection before performing maintenance or repairs.</li> <li>○ Create a step-by-step checklist covering electrical, mechanical, and operational aspects of the units to identify potential issues efficiently.</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                    |
| 8 – 11 | <ul style="list-style-type: none"> <li>● Perform troubleshooting of domestic refrigeration and window-type air-conditioning units.</li> </ul> | <ul style="list-style-type: none"> <li>● <b>Fault Diagnosis Drill</b> <ul style="list-style-type: none"> <li>○ Learners identify and diagnose common operational problems in domestic refrigerators and window-type AC units, such as cooling issues, leaks, or abnormal noises.</li> <li>○ Inspect the unit, test electrical and mechanical components, and determine the root cause of the malfunction.</li> </ul> </li> <li>● <b>Problem-Solving Simulation</b> <ul style="list-style-type: none"> <li>○ Learners simulate troubleshooting scenarios to practice correcting faults safely, including replacing or repairing defective components.</li> <li>○ Follow a systematic troubleshooting sequence using the checklist, perform necessary repairs or adjustments, and verify the unit operates correctly.</li> </ul> </li> </ul> |



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| <b>Suggested Performance Tasks</b> | <ul style="list-style-type: none"><li>• <b>Individual Performance Task - Fault Diagnosis Challenge</b><ul style="list-style-type: none"><li>○ Each learner independently troubleshoots a malfunctioning domestic refrigerator or window-type AC unit, identifying electrical, mechanical, or operational faults while following safety protocols.</li><li>○ Inspect the unit systematically, test components, identify the root cause of the problem, and record findings safely.</li></ul></li><li>• <b>Group Performance Task - Collaborative Troubleshooting Exercise</b><ul style="list-style-type: none"><li>○ In groups, learners work together to troubleshoot multiple refrigeration and AC units, sharing responsibilities such as checking circuits, testing compressors, and verifying airflow, ensuring safe procedures throughout.</li><li>○ Divide tasks among team members to diagnose all system components, identify faults, and recommend corrective actions while observing all safety precautions.</li></ul></li></ul> |
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**3<sup>rd</sup> TERM**

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| <b>Performance Standard</b> | The learner performs repairing domestic refrigeration and window-type air-conditioning units with safety precautions. |
| <b>Content Standard</b>     | Understands the concepts and principles of repairing domestic refrigeration and window-type air-conditioning units.   |

| <b>Week</b>  | <b>Learning Competency</b>                                                                                                                                            | <b>Suggested Activities</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| <b>1 - 4</b> | <ul style="list-style-type: none"> <li>Understands the concepts and principles of repairing domestic refrigeration and window-type air-conditioning units.</li> </ul> | <ul style="list-style-type: none"> <li><b>Component Repair Analysis</b> <ul style="list-style-type: none"> <li>Learners study the repair procedures for common components of domestic refrigerators and window-type AC units, such as compressors, thermostats, fans, and coils.</li> <li>Examine sample components, identify possible faults, and explain step-by-step repair methods and safety precautions.</li> </ul> </li> <li><b>Repair Procedure Demonstration</b> <ul style="list-style-type: none"> <li>Learners observe and discuss actual repair techniques for refrigeration and AC units, including replacing faulty parts and restoring proper operation.</li> <li>Watch a guided demonstration of repairs, take notes on procedures and tools used, and explain the sequence</li> </ul> </li> </ul> |

| Week                               | Learning Competency                                                                                                                                                                                                                                                                                                                                                             | Suggested Activities                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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|                                    |                                                                                                                                                                                                                                                                                                                                                                                 | of steps required for safe and effective repair.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>5 - 11</b>                      | <ul style="list-style-type: none"> <li>Perform repairing of domestic refrigeration and window-type air-conditioning units.</li> </ul>                                                                                                                                                                                                                                           | <ul style="list-style-type: none"> <li><b>Component Replacement Practice</b> <ul style="list-style-type: none"> <li>Learners practice repairing defective components such as thermostats, compressors, fans, and electrical connections in refrigeration and window-type AC units.</li> <li>Identify the faulty component, safely remove it, replace it with a new or repaired part, and reassemble the unit correctly.</li> </ul> </li> <li><b>Functional Restoration Drill</b> <ul style="list-style-type: none"> <li>Learners perform end-to-end repair procedures, ensuring the refrigeration or AC unit is restored to proper operation and tested for safety and efficiency.</li> <li>Follow a step-by-step repair sequence, test all functions, and verify that cooling, airflow, and electrical systems operate correctly.</li> </ul> </li> </ul> |
| <b>Suggested Performance Tasks</b> | <ul style="list-style-type: none"> <li><b>Individual Performance Task - Component Repair Challenge</b> <ul style="list-style-type: none"> <li>Each learner independently repairs a defective component (e.g., thermostat, fan, compressor, or electrical connection) in a domestic refrigerator or window-type AC unit while following safety protocols.</li> </ul> </li> </ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

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|  | <ul style="list-style-type: none"><li>○ Identify the faulty component, perform the repair or replacement safely, and test the unit to ensure proper operation.</li><li>● <b>Group Performance Task - Full-System Repair Project</b><ul style="list-style-type: none"><li>○ In groups, learners collaboratively repair multiple refrigeration or AC units, dividing tasks for diagnosing faults, replacing defective parts, and restoring full functionality safely.</li><li>○ Work as a team to troubleshoot, repair, and test each unit, ensuring all electrical and mechanical systems are safely restored to proper operation.</li></ul></li></ul> |
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